



**HVAC TENDER DOCUMENT – FOR ASSEMBLY LINE
VENTILATION**

desired position. The position of the handle of Damper operating mechanism shall be clearly visible and it shall indicate the position of the damper in duct. Dampers, splitters and their operating mechanism shall be fabricated of GS sheets of 16 SWG and shall be easily accessible through suitable access doors in the ducts.

Dampers shall be installed in duct at all required locations such as chutes, branches etc.

LOUVERS

Louvers complete with bird screen etc., shall be measured from the approved drawings / as built drawing on the basis of core area (excluding margin flanges) and paid per unit area. Louvers in supply and exhaust fan units are part of fan units and no separate payment will be made for the same. The louvers shall be made of extruded aluminium power coated, min. 2mm thick frame & min. 1.5mm thick fins within it.

Any supports, nuts, bolts, anchors, fasteners, gasket, tools, gadgets etc. required for successful operation and acceptance shall be provided at no extra cost to the owner.

A flanged frame using RS sections shall be provided on front face to conceal the gap between the louvers and the adjoining wall face. Corners of frame shall be welded. The frame shall be made structurally rigid.

Louvers made from extruded aluminum section shall be in modular panel form for ease of handling. These shall be free from waves and buckles. Vertical blades shall be truly vertical and horizontal blades shall be truly horizontal. Butt joints in blades shall not be accepted.

Additional intermediate equally spaced supports and stiffeners shall be provided to prevent sagging/vibrating of the louvers, at not more than 750mm centers where the louver's length is longer than 750mm.

A bird wire screen made of 12 mm mesh in 1.6 mm steel wire held in angle or channel frame shall be fixed to the rear face of the louver frame by screens.

APPARATUS AND EQUIPMENT CONNECTIONS

Equipment such as air handling units shall be connected to the duct by means of double canvas sleeve of 15 ounce, woven asbestos cloth connection of at least 150mm long.

Duct sleeves made of 20 SWG thick galvanized sheet shall be used for ducts passing through load bearing walls or partitions. Sleeves shall provide 25 mm clearance all around as per duct or insulated duct. The space between sleeve and duct shall be packed with twisted asbestos.

All the sheet metal plenums required to confine the flow of air through filters and fans, shall be fabricated out of 18 SWG galvanized sheet steel, and suitably braced as required. Suitable access doors shall be provided for plenums.

JET NOZZLE

Jet nozzles shall be suitable for installation on circular ducts. The discharge nozzle and face cover ring shall be made of aluminium, the mounting plate for the eyeball is aluminium or sheet steel, The duct connection element and duct connection pieces shall be of galvanised sheet steel. Jet nozzle, cover ring, eyeball, mounting plates shall be finish powder coated, resistant to saturate environment for minimum 100 hours without deterioration. Jet nozzle shall be available with rear mounted perforated sheet steel plate for flow rate control.

Jet nozzles shall be suitable for long throw distances with optimum acoustic properties, preferably used for cooling in critical areas. Jet nozzles shall provide the adjustment facility manual or automatic using an electric actuator which allows variation in discharge angle to compensate for changing temperature differences, adjustment angular range 30° upwards to 30° downwards.

Jet nozzle shall be selected for minimum pressure drop and sound level not exceeding 35NC, contractor shall submit the selection curves of product ratings from the manufacturer.

12 INSULATION

The Inside surface of the first 3.0 meters of supply and return air-conditioning duct from AHU/CSU/ TFA/ FCU on which the acoustic lining is to be provided shall be thoroughly cleaned and rendered free from all dust and grease. The portion of supply air duct which is acoustically (internal) insulated need not be insulated thermally (external).

Noted



Project Name:

PERKINS INDIA PVT. LTD.

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Good quality adhesive as per manufacturer recommendation should be used for pasting 100% surface area of insulation to the duct surface without leaving any air bubble or sag between two surfaces. The joints shall be paste along the thickness of the insulating material. The longitudinal & radial joints should have overlapping 3" strip of same material with min 3mm thickness. Mechanical fasteners shall be used to ensure adherence of duct insulation to surface of the ducts.

The insulation material shall be open cell nitrile rubber and the thickness of insulation shall be 10mm for FCU duct and 15mm for the AHU/ TFA duct

Duct thermal Insulation using closed cell nitrile rubber

DETAIL SPECIFICATION OF INSULATION IS TO BE GIVEN.

The outer surface of the supply and return air-conditioning duct on which the insulation is to be provided shall be thoroughly cleaned and rendered free from all dust and grease.

Good quality adhesive as per manufacturer recommendation should be used for pasting 100% surface area of insulation to the duct surface without leaving any air bubble or sag between two surfaces. The joints shall be paste along the thickness of the insulating material. The longitudinal & radial joints should have overlapping 2" strip of same material with min 3mm thickness.

The insulation material shall be aluminium foil faced closed cell nitrile rubber and the thickness of insulation shall be as per following table or identified in the schedule of quantity:

Supply Duct in conditioned space (return above FC)	- 19 mm thick
Return Duct in conditioned space / Supply Duct Evap ventilation.	- 13 mm thick
Supply Duct in exposed / non-conditioned space	- 25 mm thick
Return Duct in exposed / non-conditioned space	- 19 mm thick

Refrigerant/ Drainpipe Insulation using closed cell nitrile rubber

The outer surface of the pipes on which the insulation is to be provided shall be thoroughly cleaned and rendered free from all dust and grease.

Good quality adhesive as per manufacturer recommendation should be used for pasting 100% surface area of insulation to the pipe surface without leaving any air bubble or sag between two surfaces. The joints shall be paste along the thickness of the insulating material. The longitudinal & radial joints should have overlapping 2" strip of same material with min 3mm thickness.

The insulation material shall be closed cell nitrile rubber covered with 6-7 mil minimum thickness and 200 gram/sqmt $\pm 10\%$ interwoven glass fabric for UV and mechanical protection and the thickness of insulation shall be as identified in the schedule of quantity. All valves, fittings etc. shall be insulated to the same thickness and in the same manner as for the respective piping, taking care to allow operation of valves without damaging the insulation.

Noted



----- **END OF SECTION** -----